1. Program — prac1.l

```
%{
#include <stdio.h>
%}

%option noyywrap

%%
"int"|"float"|"char"|"if"|"else"|"while"|"for"|"return"  { printf("KEYWORD : %s\n", yytext); }
[A-Za-z_][A-Za-z0-9_]*   { printf("IDENTIFIER : %s\n", yytext); }
[0-9]+                    { printf("NUMBER     : %s\n", yytext); }
"+"|"-"|"*"|"/"|"="|"<"|">" { printf("OPERATOR   : %s\n", yytext); }
[(){};,]                  { printf("DELIMITER  : %s\n", yytext); }
[ \t\n]+                  { }
.                         { printf("UNKNOWN    : %s\n", yytext); }
%%

int main(int argc, char **argv) {
    if (argc > 1) yyin = fopen(argv[1], "r");
    yylex();
    return 0;
}
```

2. Input File — input1.c

```
int sum = 0;
while (sum < 10) {
    sum = sum + 2;
}
```

3. Commands

```
flex prac1.l
gcc lex.yy.c -o prac1
./prac1 input1.c
```

4. Output

```
total lexical errors: 1
nirma@nirma-44:~/24bce527$ gedit prac1.l
nirma@nirma-44:~/24bce527$ gedit input1.c
nirma@nirma-44:~/24bce527$ flex prac1.l
nirma@nirma-44:~/24bce527$ gcc lex.yy.c -o prac1
nirma@nirma-44:~/24bce527$ ./prac1 input1.c
KEYWORD    : int
IDENTIFIER : sum
OPERATOR   : =
NUMBER     : 0
DELIMITER  : ;
KEYWORD    : while
DELIMITER  : (
IDENTIFIER : sum
OPERATOR   : <
NUMBER     : 10
DELIMITER  : )
DELIMITER  : {
IDENTIFIER : sum
OPERATOR   : =
IDENTIFIER : sum
OPERATOR   : +
NUMBER     : 2
DELIMITER  : ;
DELIMITER  : }
nirma@nirma-44:~/24bce527$
```